



FINAL EXAMINATION

Course: CS251 Logical Structures

Time: 75 minutes

Term: III – Academic year: 2019-2020

Lecturer(s): Nguyen Phuc Son

Student name:

Student ID:

(Notes: Neither books nor laptops, phones allowed)

<Followings are the questions and/or requirements>

The test has **4 problems**.

Question 1) Draw **parse trees** and identify the **free and bound variables** for the formulas below

(a) $(\forall x)(\forall y)(R(x) \wedge S(y) \longrightarrow S(x) \wedge R(y)) \wedge (y = c)$

(b) $\exists z(S(z, x) \vee (\forall x(\neg Q(z, x) \wedge Q(x, x))))$

Question 2) Use the **tableau algorithm** to determine whether the following semantic entailment holds or not:

$$\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \longrightarrow \neg R(x, z)) \models \forall x \neg R(x, x)$$

Question 3) A formula ϕ is given by the truth table below:

p	q	r	ϕ
1	1	1	1
1	1	0	0
1	0	1	1
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	0

(a) Provide the Disjunctive Normal Form (DNF) of the formula ϕ

(b) Provide the Conjunctive Normal Form (CNF) of the formula ϕ

Question 4) Either prove or disprove the following semantic equivalences (a counterexample is good enough for disproving an semantic entailment):

- (a) $\left((\forall x P(x)) \vee (\forall x Q(x)) \right) \equiv \left(\forall x (P(x) \vee Q(x)) \right)$
- (b) $\left((\exists x P(x)) \vee (\exists x Q(x)) \right) \equiv \left(\exists x (P(x) \vee Q(x)) \right)$

THE END